

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO2 ASSESSMENT TOOL 1 – Scenario-Based Requirement Analysis

Course Code:	CSA1008	Course Title:	Software Engineering
CO Assessed:	CO2	Assessment:	Assessment Tool 2 – Scenario-Based Requirement Analysis
Weightage:	20%	Total Marks:	25
CO2 Statement:	Perform detailed requirements analysis, design scalable architectures, and prioritize features with a focus on cross-disciplinary skills		
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Date:	06.08.2026	Duration:	60 minutes

Code of Conduct

I ARTHI A (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student:

Arthi. A.

Annexure A – Scenario-Based Requirement Analysis

Instructions to Students:

Each student will be assigned an individual software project problem statement. Analyze the given scenario and prepare a complete Software Requirement Analysis document by following the steps below.

Question

Based on the Case Study 36 – SDG 9: Smart Construction Project Management System. Construction companies manage multiple projects simultaneously using spreadsheets and manual reporting. Delays in material procurement, poor communication, and lack of progress tracking often result in cost overruns. The company plans to adopt a centralized software solution for project planning, resource management, and progress monitoring. As a software engineer, analyze the above problem and propose a suitable software engineering solution., perform a Scenario-Based Requirement Analysis and prepare a Software Requirement Specification (SRS) document. Your analysis should include the following:

1. Study and Analyze the Scenario

- Carefully understand the given problem statement.
- Identify the business domain, stakeholders, existing challenges, and system objectives.

2. Identify Functional and Non-Functional Requirements

- List all functional requirements required by the proposed system.
- Identify suitable non-functional requirements such as performance, security, reliability, usability, scalability, maintainability, and availability.

3. Identify Actors and Use Cases

- Identify all primary and secondary actors interacting with the system.
- List the major use cases associated with each actor.

- Draw a Use Case Diagram representing the interactions between actors and the system.
- 4. **Prepare the Software Requirement Specification (SRS)**
Develop an SRS document containing:
 - Introduction
 - Problem Description
 - Scope of the System
 - Functional Requirements
 - Non-Functional Requirements
 - Use Case Descriptions
 - Data Requirements
 - External Interface Requirements
 - System Features
- 5. **Identify Assumptions and Constraints**
 - List the assumptions considered while developing the system.
 - Identify technical, operational, economic, legal, and time-related constraints affecting the project.
- 6. **Validate Requirement Completeness and Consistency**
 - Verify whether all stakeholder requirements have been addressed.
 - Check for ambiguity, redundancy, inconsistency, and missing requirements.
 - Suggest improvements to enhance the quality and completeness of the requirement specification.

Rubrics for Calculation

Criteria	Excellent (4 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Scenario Understanding and Problem Analysis	Demonstrates thorough understanding of the scenario, stakeholders, objectives, and business context with clear analysis.	Demonstrates good understanding with minor omissions.	Basic understanding with limited analysis.	Poor understanding; major aspects missing or incorrect.
Functional and Non-Functional Requirement Identification	Clearly identifies comprehensive, accurate, and well-classified functional and non-functional requirements.	Identifies most requirements with minor classification errors.	Identifies only basic requirements; several omissions.	Requirements are incomplete, inaccurate, or poorly classified.
Actor Identification and Use Case Modeling	Correctly identifies all relevant actors and use cases; use case diagram is complete, accurate, and follows UML standards.	Most actors and use cases identified; diagram has minor errors.	Limited actors/use cases identified; diagram partially correct.	Actors/use cases missing or incorrect; diagram absent or inaccurate.

Software Requirement Specification (SRS) Documentation	SRS is well-structured, complete, professionally organized, and includes all required sections.	SRS is organized with minor missing sections or formatting issues.	SRS includes only essential sections with limited detail.	SRS is incomplete, poorly organized, or lacks essential components.
Assumptions, Constraints, and Requirement Validation	Clearly identifies realistic assumptions and constraints; validates requirements for completeness, consistency, and ambiguity with appropriate justification.	Identifies most assumptions and constraints; validation is adequate.	Limited assumptions/constraints; validation is superficial.	Assumptions, constraints, and validation are missing or incorrect.

Criteria	Mark
Scenario Understanding and Problem Analysis	/5
Functional and Non-Functional Requirement Identification	/5
Actor Identification and Use Case Modeling	/5
Software Requirement Specification (SRS) Documentation	/5
Assumptions, Constraints, and Requirement Validation	/5
TOTAL	/5

1. Scenario Understanding & Problem Analysis

Business Domain

Construction Project Management

Problem Summary

Construction companies managing multiple simultaneous projects currently rely on spreadsheets and manual, informal reporting such as phone calls, messages, and paper logs to track progress and coordinate material procurement. This produces three compounding problems. First, a reporting lag between what is actually happening on-site and what management knows, often one to two days or more. Second, untracked, informal requisition requests that are easy to lose, duplicate, or delay. Third, client-facing reporting that depends on the contractor's manual summary rather than verifiable, timestamped data, which is a common source of payment disputes. Together these drive the two outcomes stated in the original problem brief: cost overruns and schedule delays.

System Objectives

- Centralize project planning, resource management, and progress monitoring into a single platform
- Eliminate manual/spreadsheet-based reporting delays
- Enable real-time communication between site engineers, project managers, procurement, and clients
- Reduce cost overruns caused by late material ordering
- Provide transparent, verified progress visibility to clients tied to payment milestones

Stakeholders

Stakeholder	Business Context
Site Engineer	Works on-site, currently reports progress manually (paper/spreadsheet); needs a fast way to log data and flag material shortages
Project Manager	Oversees multiple projects from the office; currently has no consolidated real-time view, relies on delayed manual reports
Procurement Team	Currently reacts to requisitions late because they arrive via informal channels (calls, emails); needs a structured, trackable request-to-order pipeline
Client	Currently receives progress updates infrequently and manually; wants transparent, verifiable milestone tracking before releasing payments
Company Admin/Owner	Needs a portfolio-level view across all active projects to spot risk early
Suppliers/Vendors	External parties who fulfill purchase orders; interact indirectly through procurement's PO tracking

Existing Challenges

- No single source of truth: the same project's status may differ depending on who you ask and when, which is the root cause of most downstream disputes
- Reporting lag of one to two or more days between site events and management awareness, directly delaying corrective action
- Material shortages discovered reactively, when work stops, rather than proactively, when the shortage is first identified
- Low client trust due to unverifiable, self-reported progress claims tied to payment requests
- No structured, timestamped approval trail, meaning decisions and their rationale are not consistently recorded, complicating audits or disputes

2. Functional Requirements (12)

ID	Requirement
FR-1	The system shall allow Site Engineers to log daily progress updates with photos and notes via a mobile app
FR-2	The system shall allow Site Engineers to update task completion status against the project schedule
FR-3	The system shall allow Site Engineers to raise material requisition requests from the field
FR-4	The system shall allow Project Managers to view a consolidated dashboard of progress across all assigned projects
FR-5	The system shall allow Project Managers to approve or reject material requisitions
FR-6	The system shall allow Project Managers to view and adjust project schedules based on actual progress
FR-7	The system shall allow Project Managers to approve/verify site-reported milestones
FR-8	The system shall allow Procurement Officers to convert approved requisitions into purchase orders
FR-9	The system shall allow Procurement Officers to track and update purchase order delivery status
FR-10	The system shall allow Clients to view real-time, verified milestone progress via a client portal
FR-11	The system shall allow Clients to approve payments tied to verified milestones
FR-12	The system shall send notifications to relevant users when an action is pending (approval, delayed PO, etc.)

Non-Functional Requirements (10)

ID	Category	Requirement
NFR-1	Performance	Progress updates submitted from the mobile app shall reflect on the PM dashboard within 5 minutes
NFR-2	Usability	The mobile interface shall be usable in low-connectivity site environments with minimal data entry steps
NFR-3	Security	All user data shall be protected via role-based access control (Site Engineer, PM, Procurement, Client, Admin)

NFR-4	Security	Client payment approvals shall require authenticated login and be logged with timestamps
NFR-5	Reliability	The system shall maintain 99.5% uptime during business hours
NFR-6	Scalability	The system shall support multiple concurrent projects and a growing number of users without performance degradation
NFR-7	Maintainability	The system shall be built with modular components to allow independent updates to mobile, web dashboard, and portal modules
NFR-8	Availability	Core progress-logging and requisition features shall remain available even during partial network interruptions (offline caching with sync-on-reconnect)
NFR-9	Interoperability	The system shall support data export (CSV/PDF) for reporting and audit purposes
NFR-10	Compliance	The system shall retain audit trails of all approvals (requisitions, milestones, payments) for a minimum of 3 years

3. Actors & Use Cases

Primary Actors

- Site Engineer
- Project Manager
- Procurement Officer
- Client

Secondary Actors

- Company Admin (oversight, not direct daily use)
- Supplier/Vendor (external, receives POs — not a direct system user in this scope)

Use Cases per Actor

Site Engineer:

- Log Daily Progress
- Update Task Status
- Raise Material Requisition
- View Assigned Schedule

Project Manager:

- View Consolidated Dashboard
- Approve/Reject Requisition
- Adjust Project Schedule
- Approve Milestone
- View Portfolio Summary (multi-project)

Procurement Officer:

- Convert Requisition to Purchase Order
- Track PO Delivery Status

- Update PO Status

Client:

- View Milestone Progress
- Approve Payment

Company Admin (secondary):

- View Cross-Project Summary Report

Use Case Diagram:



Structure: 4 actors (Site Engineer, Project Manager, Procurement Officer, Client) connected via association lines to their respective use cases, all contained within a system boundary box labeled "Smart Construction Project Management System".

4. Software Requirement Specification (SRS)

4.1 Introduction

This SRS defines the requirements for the Smart Construction Project Management System, a centralized platform designed to replace manual, spreadsheet-based project tracking with real-time digital workflows connecting site engineers, project managers, procurement, and clients.

4.2 Problem Description

Construction firms managing multiple simultaneous projects face delays and cost overruns due to fragmented, manual communication — particularly around material procurement and progress reporting. This system addresses that gap through role-specific interfaces feeding into one shared data source.

4.3 Scope of the System

The system covers: mobile progress/requisition logging (Site Engineer), web-based schedule and approval management (Project Manager), purchase order tracking (Procurement), and a client-facing milestone/payment portal. It does not cover accounting/ERP integration, HR management, or direct supplier-side ordering systems (out of scope for this phase).

4.4 Functional Requirements

See Section 2 (FR-1 to FR-12) above.

4.5 Non-Functional Requirements

See Section 2 (NFR-1 to NFR-10) above.

4.6 Use Case Descriptions

UC-1: Log Daily Progress

- Actor: Site Engineer
- Precondition: User is authenticated on the mobile app
- Main Flow: Engineer opens app → selects project/task → enters progress notes and photo → submits

- Postcondition: Progress entry stored, visible on PM dashboard

UC-2: Approve Material Requisition

- Actor: Project Manager
- Precondition: A requisition exists in "Pending" status
- Main Flow: PM opens requisition queue → reviews request details → approves or rejects with comment
- Postcondition: Status updated; Site Engineer and Procurement notified

UC-3: Approve Payment

- Actor: Client
- Precondition: Milestone status is "Verified"
- Main Flow: Client opens portal → reviews milestone → approves payment
- Postcondition: Payment status set to "Approved," logged with timestamp

4.7 Data Requirements

- User data: name, role, contact, project assignment
- Progress logs: timestamp, task ID, status, photo reference, notes
- Requisitions: material, quantity, urgency, status, linked project/task
- Purchase orders: PO number, linked requisition ID, delivery status, supplier reference
- Milestones: linked tasks, verification status, associated payment amount
- Notifications: recipient, trigger event, timestamp, read/unread status

4.8 External Interface Requirements

- Mobile App UI — for Site Engineers (Android/iOS)
- Web Dashboard UI — for Project Managers (browser-based)
- Client Portal UI — for Clients (browser-based, restricted view)
- Procurement Queue UI — for Procurement Officers (browser-based)
- Notification Interface — push notifications (mobile) and email/in-app alerts (web)
- No direct third-party hardware interfaces required in this phase

4.9 System Features

- Role-based dashboards per user type
- Real-time data sync across mobile and web
- Approval workflows with audit trail
- Milestone-linked payment gating (client can't pay for unverified work)
- Notification engine for pending actions

5. Assumptions and Constraints

Assumptions

All users have access to a smartphone, in the case of Site Engineers, or a computer and browser, in the case of the Project Manager, Procurement, and Client roles. Construction sites have at least intermittent internet connectivity sufficient for periodic sync. Project schedules are initially set up by the Project Manager before Site Engineers begin logging progress against them. Payment amounts per milestone are predefined in the underlying project contract, not calculated by the system itself. Users are onboarded with correctly assigned roles, and the system does not currently handle self-service role requests.

Constraints

Type	Constraint
Technical	Mobile app must function with limited/offline connectivity (site environments)
Operational	Site Engineers may have limited technical literacy — UI must be simple and minimal-input
Economic	Solution must be cost-effective to deploy across multiple simultaneous project sites
Legal	Payment approval logs must be retained for audit/compliance purposes (see NFR-10)
Time	Initial rollout (Sprint 1 scope) is limited to core field-to-office data flow; full feature set is phased across multiple sprints

6. Requirement Validation

6.1 Completeness Check

All six stakeholders have at least one functional requirement directly addressing their core need. All four primary actors have associated use cases, and all use cases trace back to at least one functional requirement. Every functional requirement maps to at least one Agile user story in the accompanying Sprint Planning document, confirming end-to-end traceability from requirement to delivery.

6.2 Consistency and Ambiguity Check

Requirements were cross-checked pairwise for conflicts, and none were found. A specific check was made between FR-7, where the Project Manager approves a milestone, and FR-11, where the Client approves payment, to confirm they represent sequential, non-overlapping authority. The Project Manager verifies that work was actually completed, and only after that verification can the Client approve the associated payment. This avoids ambiguity about who has final authority over what, and prevents a scenario where a client could approve payment for unverified work.

6.3 Redundancy Check

No duplicate or overlapping requirements were identified across the twelve functional requirements. Where two requirements involve the same actor, such as FR-5 and FR-7, both involving the Project Manager, they govern clearly distinct objects, requisitions versus milestones.

6.4 Missing Requirements Identified and Resolved

During validation, notification handling was initially absent from the requirement set. This was identified as a gap because several approval-based use cases depend on the relevant party being alerted promptly. Without it, the system would technically function but silently fail to deliver on the objective of reducing reporting lag. FR-12 and NFR-8, offline handling, were added specifically to close this gap.

6.5 Suggested Improvements

Add multi-language support as a future requirement, since site engineers across regions may not share a common working language. Add a dispute or rejection-reason log for payment approvals, to support resolution when a client and contractor disagree on verified progress. Add role-based reporting exports, such as PM-only cost reports and Admin-only portfolio risk reports, as a future enhancement beyond current scope. Consider a formal SLA-tracking requirement for purchase order delivery times, to strengthen the procurement-delay objective stated in the original problem brief.